9 Chemical Kinetics

SOLUTION (QR CODE)

10. Disappearance rate of $N_2 = -\frac{d[N_2]}{dt}$

Disappearance rate of $H_2 = -\frac{d[H_2]}{dt}$

Appearance rate of $NH_3 = \frac{d[NH_3]}{dt}$

$$-\frac{1}{1}\frac{d[N_2]}{dt} = -\frac{1}{3}\frac{d[H_2]}{dt} = \frac{1}{2}\frac{d[NH_3]}{dt} = \text{RoR}$$

Disappearance mei minus sign lgane ka or appearance mei plus sign lagane ka

11. (1) According to the given reaction:

$$-\frac{d[N_2]}{dt} = -\frac{1}{3}\frac{d[H_2]}{dt} = \frac{1}{2}\frac{d[NH_3]}{dt}$$

= Rate of reaction

Given: $-\frac{d[N_2]}{dt} = x$ mol/L/s

Rate of reaction = x mol/L/s

(2) $-\frac{d[N_2]}{dt} = -\frac{1}{3}\frac{d[H_2]}{dt}$, $-\frac{d[H_2]}{dt} = 3x$

(3) $-\frac{d[N_2]}{dt} = \frac{1}{2}\frac{d[NH_3]}{dt}$, $\frac{d[NH_3]}{dt} = 2x$

(4) $\frac{1}{2}\frac{d[NH_3]}{dt}$ = Rate of reaction

Rate of reaction = x

13. (2.67 M/min) According to the given reaction,

$$-\frac{1}{3}\frac{d[C_2H_6O]}{dt} = \frac{1}{2}\frac{d[Cr_2(SO_4)_3]}{dt}$$

$$\frac{d[Cr_2(SO_4)_3]}{dt} = \frac{2}{3} \times \frac{d[C_2H_6O]}{dt}$$

$$= \frac{2}{3} \times 4 \text{ M/min} = 2.67 \text{ M/min}$$

15. (A) Given:

$$-\frac{1}{3}\frac{d[Br_2]}{dt} = -\frac{1}{6}\frac{d[OH^-]}{dt} = \frac{1}{5}\frac{d[Br^-]}{dt} = \frac{1}{1}\frac{d[BrO_3^-]}{dt}$$

= ROR

$$\frac{d[BrO_3^-]}{dt} = 1.5 \times 10^{-2} \text{ m/sec}$$

(A) $(ROD)_{Br_2} = -\frac{d[Br_2]}{dt} = 3\frac{d[BrO_3^-]}{dt}$

$= 1.5 \times 10^{-2} \times 3 = 4.5 \times 10^{-2}$ M/sec

(B) $\frac{d[Br^-]}{dt} = (ROA)_{Br^-} = 5\frac{d[BrO_3^-]}{dt} = 5 \times 1.5 \times 10^{-2}$

$= 7.5 \times 10^{-2}$ M/sec

sara khel minus plus ka hai

20. (B) $-\frac{1}{n}\frac{d[x]}{dt}$

Species

reactant

Stoichiometric coefficient

According to this A, B and E are reactant- with Stoichiometric coefficient 2, 4 and 5 respectively

$+\frac{1}{n}\frac{d[x]}{dt}$

Species

Product

stoichiometric coefficient

According to this 'C' and 'D' are products with Stoichiometric coefficient 3 and 3 respectively-

So the net reaction is-

$2A + 4B + 5E \rightarrow 3C + 3D$

22. $0.96 = 2 \times 0.48 = 2\log 3 = \log 3^2 = \log 9$

$0.60 = 2 \times 0.30 = 2\log 2 = \log 2^2 = \log 4$

So the given equation is converted into -

$$\log 9 + \log\left[\frac{-d[A]}{dt}\right] = \log\left[\frac{+d[B]}{dt}\right] + \log 4$$

$$-9\frac{d[A]}{dt} = 4\frac{d[B]}{dt}, \quad -\frac{1}{4}\frac{d[A]}{dt} = \frac{1}{9}\frac{d[B]}{dt}$$

So the equation is $4A \rightarrow 9B$

34. $R = k[A][B]^2$

According to the given information,

$2R = k[2A][B]^2$

$R' = k[2A][2B]^2$

On comparing both equations, the rate will increase by 8 times with respect to the original rate.

36. Rate = k[A][B]
Rate' = 4Rate
Which means that concentrations of both A and B will be doubled.
As concentration becomes double, volume gets reduced to half.

38. (B) First write the general expression of rate law in terms of x and y:

$R = k[A]^x [B]^y$

Here, x and y are orders of reaction with respect to [A] and [B] respectively.

Divide experiments (1) and (2):

$0.10 = [0.30]^x [0.30]^y$, $0.40 = [0.30]^x [0.60]^y$

On solving, $0.25 = (0.5)^y$, y = 2

Divide experiments (1) and (3):

$0.10 = [0.30]^x [0.30]^y$

$0.20 = [0.60]^x [0.30]^y$

$0.5 = (0.5)^x$

On solving, x = 1

Hence, the correct expression of rate law is:

$Rate = k[A][B]^2$

42. (D) Let n be the order of reaction.

$R = k\,(Reactants)^n$ (*i*)

According to the given information:

$8R = k(2.reactants)^n$ (*ii*)

On dividing equations (ii) and (i)

$(2)^3 = k(2)^n$

So, n = 3

When concentration of G doubles keeping H constant, rate also becomes double,

$Rate = k[G][H]^2$

So, overall order of reaction is 3.

Bhut hi asaan hai

45. (B) Let rate = $K[A]^m [B]^n$

Using the date from exp 1 and exp 2 we get n = 2

Using the date from exp 1 and exp 3 we get m = 1

So, rate = $K[A][B]^2$

Now using data from exp 2 and exp 4 we get,

$$\left.\begin{aligned} 2.4 \times 10^{-2} &= K[0.1][0.2]^2 \\ 7.2 \times 10^{-2} &= K[X][0.2]^2 \end{aligned}\right\} X = 0.3$$

Now using data from exp 1 and exp 5 we get,

$$\left.\begin{aligned} 6 \times 10^{-3} &= K[0.1][0.1]^2 \\ 2.88 \times 10^{-1} &= K[0.3][Y]^2 \end{aligned}\right\} Y = 0.4$$

table par dhyan lagao pura apna

48. The general formula to determine units of rate constant is:

$(conc)^{1-n}\ time^{-1}$

Here, n = 2

Units of rate constant = $(conc)^{1-2}\ (sec)^{-1}$

$= Mol^{-1}\ L\ sec^{-1}$

55. $\frac{A_o}{2k} = 50$

$k = \frac{A_o}{100}$, 80% completion means $A_t = 0.2\ A_o$

$A_o - 0.2A_o = kt_{80\%}$

$0.8\ A_o = \frac{A_o}{100} t_{80\%}$, $t_{80\%} = 80$ s

66. This is a zero order kinetics by seeing
- → surface catalysed mechanism
- → Unit of k.

Rate of reaction = $K[NH_3]^0 = 2.5 \times 10^{-4}$

$$-\frac{1}{2}\frac{d[NH_3]}{dt} = \frac{d[N_2]}{dt} = \frac{1}{3}\frac{d[H_2]}{dt} = \text{Rate of reaction}$$

$$\frac{d[N_2]}{dt} = k = 2.5 \times 10^{-4}$$

$$\frac{d[H_2]}{dt} = 3 \times 2.5 \times 10^{-4} = 7.5 \times 10^{-4}$$

balle balle ho gayi

71. For first-order reaction,

$$k = \frac{2.303}{t} \log \frac{a}{a-x}$$

When $x = \frac{40}{100} a = 0.4\ a$

t = 50 minutes (given)

$$\therefore k = \frac{2.303}{50 \text{ min}} \log \frac{a}{a - 0.4\,a} \text{ or } k$$

$$= \frac{2.303}{50 \text{ min}} \log \frac{a}{0.6} = 0.010216 \text{ min}^{-1}$$

Now, when reaction gets 80 % completed, then,

i.e., x = 0.8a

$$\therefore t = \frac{2.303}{k} \log \frac{a}{a - x}$$

$$= \frac{2.303}{0.010216 \text{ min}^{-1}} \log \frac{a}{a - 0.8a},$$

t = 157.58 min

78. According to the given graph, $t_{1/2}$ is directly proportional to $[A]_0$, hence the order of the reaction is zero order.

For zero order reaction,

$$t_{1/2} = \frac{[A]_0}{2k} \Rightarrow \text{slope} = \frac{1}{2k}$$

Given, slope = 76.92

$$\therefore \frac{1}{2k} = 76.92$$

$$\Rightarrow k = \frac{1}{76.92 \times 2}$$

Now, $[A]_t = [A]_0 - kt$

$$[A]_{10} = 2.5 - \frac{1}{(76.92 \times 2)} \times 10$$

$= 2.5 - 0.0625 = 2.435 \text{ mol L}^{-1}$

$\therefore [A]_{10} = 2.435 \text{ mol L}^{-1}$

$= 2435 \times 10^{-3} \text{ mol L}^{-1}$

82. For first order reaction,

$$k = \frac{2.303}{t} \log \frac{[A]_0}{[A]}$$

Calculate time required for 99.9% completion of reaction is:

$$k = \frac{2.303}{t} \log \frac{100}{100 - 99.9}$$

$$k = \frac{2.303}{t} \log \frac{100}{0.1} = \frac{2.303}{t} \log 1000$$

$$k = \frac{2.303}{t} \times 3 \; (\because \log 1000 = 3)$$

$$k = \frac{6.909}{k} \quad \text{...(1)}$$

Half-life for first-order reaction is given by:

$t_{1/2} = 0.693/k$ ...(2)

Now compare both equations (1) and (2):

$$\frac{t}{t_{1/2}} = \frac{6.909}{k} \times \frac{k}{0.693}$$

$$\frac{t}{t_{1/2}} = \frac{6.909}{0.693} = 10, \quad t = 10 \times t_{1/2}$$

90. The formula to calculate rate constant is:

$$k = \frac{2.303}{t} \log \frac{r_\infty - r_o}{r_\infty - r_t}$$

Substitute the given values:

$$5.36 \times 10^{-5} \text{ s}^{-1} = \frac{2.303}{1435} \log \left(\frac{r_\infty - 34.50}{r_\infty - 31.10} \right)$$

$r_\infty = -11$

thoda tedha hai

96. (C) The Arrhenius equation applies to both elementary and complex reactions. The unit of the pre-exponential factor (A) is the same as the unit of the rate constant (k). The rate of a reaction is higher when the activation energy (E_a) is lower. Additionally, A and E_a are considered temperature-independent.

100. (D) Arrhenius equation in terms of lnk is:

$$\ln k = \ln A - \frac{E_a}{RT}$$

In terms of k_1 and k_2:

$$\ln \frac{k_2}{k_1} = \frac{E_a}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$$

On comparing with straight line equation,

$$-\frac{E_a}{R} (\text{slope}) = -4606$$

Now, substitute the values in above expression:

$$\ln \frac{k_2}{10^{-5}} = 4606 \left(\frac{1}{400} - \frac{1}{500} \right)$$

$$\ln \frac{k_2}{10^{-5}} = 2.303\ , \quad \frac{k_2}{10^{-3}} = 10 \Rightarrow k_2 = 10^{-4}\ s^{-1}$$

seedha sa sawaal hai tedha mat samajhna

101. $t_{1/2} = \dfrac{0.693}{k}$

Also, $k = Ae^{-E_a/RT}$

$$k = 10^{20} \times e^{\frac{-191.48\times10^3}{8.314\times1000}} = 10^{10}\ sec.$$

$$\frac{0.693}{10^{10}} = 6.93 \times 10^{-11}$$

$= 69.3 \times 10^{-12}$ sec or 69.3 picoseconds.

kuch nhi kia bas formula lagaya hai